Road Intelligence Platform

Comprehensive Findings

Prepared for: Ryan Thomson

Prepared by: Tony Jung

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Data: 1.52M GPS breadcrumbs, 9 graders, 367 days. 1,101 fuel fill records. 3,097 road segments. 7 grader camps.

At a glance

- 24.9% fleet deadhead ratio
- MG-048 is the biggest single inefficiency
- Spare coordination is the main overlap opportunity

Executive summary

The Road Intelligence Platform baseline shows the fleet is reaching nearly every gravel segment, but doing so with material deadhead and overlap costs. The highest-value opportunities are route optimization during grading season, tighter spare deployment, and targeted investigation of MG-048.

Executive summary

Fleet deadhead	Fuel impact	Best improvement lever
24.9%	**$55k-$65k**	**Spare + route**
Active time spent traveling	Estimated annual fleet-wide fuel cost	Most actionable near-term operational fix

24.9% deadhead

The fleet spends nearly one-quarter of active time traveling between work areas. During grading season the working range is closer to 17-22%. Fleet-wide deadhead fuel is estimated at $55,000-$65,000 per year.

MG-048 is the biggest single issue

MG-048 deadheads more kilometres than it grades: 55 km of deadhead versus 36 km of grading per shift. Combined with a 23.6 L/hr burn rate, it is the most expensive inefficiency in the fleet.

Spare deployment drives overlap

93% of multi-grader overlap involves a spare grader. The core issue is not mostly territory boundaries - it is dispatch coordination and avoiding re-grading roads already completed by the primary unit.

1. Fleet deadhead analysis

Overall: 24.9%

Metric	Value
Total grading hours (8 graders excl MG-049)	5,809
Total deadhead hours	1,922
Total idle hours	1,826
Fleet deadhead ratio	24.9%
Estimated deadhead fuel cost (7 tracked graders)	$41,500
Estimated fleet-wide deadhead fuel cost (14 graders)	$55,000-$65,000

MG-049 (Division 7) is excluded from the fleet roll-up because the GPS and telematics record is anomalous.

Per-grader ranking

Rank	Grader	Division	Deadhead	Grading Hrs	Deadhead Hrs	DH Fuel
1 (worst)	MG-048	Div 5	**34.3%**	628	327	$7,872
2	MG-039	Spare	31.4%	451	206	$4,438
3	MG-050	Div 4	27.9%	1,005	388	$7,489
4	MG-051	Div 1	26.2%	660	234	$4,787
5	MG-044	Spare	22.1%	1,015	288	$6,804
6	MG-046	Div 3	21.4%	666	182	$3,866
7	MG-047	Div 6	**17.8%**	1,081	233	$4,878
8 (best)	MG-041	Spare	**17.2%**	302	63	$1,353

Seasonal pattern

Period	Shifts	Grading Hrs	Deadhead Ratio	Context
May 2025	202	748	21.7%	Spring startup, high demand
Jun 2025	192	689	18.7%	Peak season ramping up
Jul 2025	144	538	18.3%	Mid-summer maintenance
Aug 2025	164	664	17.4%	Peak efficiency
Sep 2025	163	583	16.6%	Best month - high demand, efficient routing
Oct 2025	148	563	18.3%	Season winding down
Nov 2025	137	403	33.4%	Winter transition - grading drops, travel does not
Dec 2025	140	255	53.0%	Minimal grading, mostly travel/checks
Jan 2026	157	437	28.1%	Winter operations
Feb 2026	140	443	30.7%	Pre-spring
Mar 2026	174	454	31.0%	Spring thaw beginning

Key insight

The annual 24.9% deadhead ratio is inflated by winter. The main optimization focus should be the May-October grading season, where demand is highest and absolute savings per shift are greatest.

2. MG-048 (Division 5) deep dive

MG-048 is the fleet's most expensive deadhead problem. It deadheads 55 km per shift but grades only 36 km, meaning more travel than productive work.

Metric	MG-048	Fleet Avg
Shift duration	8.9 hrs	-
Grading per shift	3.7 hrs (42%)	-
Deadhead per shift	1.9 hrs (21%)	-
Grading km per shift	36 km	-
Deadhead km per shift	55 km	-
Fuel rate	23.6 L/hr	21.1 L/hr
JD high-load %	24%	5-9%

Why MG-048 matters

Three factors stack together: Division 5 carries the heaviest gravel workload, MG-048 has the highest

measured fuel rate, and route patterns may still be improvable. That makes this unit the single richest target for operational and mechanical follow-up.

Recommended follow-up

- Investigate whether the 24% high-load profile is driven by terrain, operator technique, or equipment condition.
- Evaluate the MG-050/MG-048 swap after the high-load issue is understood.
- Use the Route Planner to test whether daily sequencing can cut MG-048 deadhead materially during grading season.

3. Route overlap analysis

2,017 of 2,542 visited gravel segments were touched by two or more graders. The headline matters, but the breakdown matters more.

The 79% number in context

Overlap Type	Segments	% of Overlap	Actionable?
Involves at least one spare grader	1,871	93%	Yes - spare coordination
2+ primary graders overlap (with spares too)	453	22%	Yes - territory + spare
Pure primary-vs-primary overlap (no spare)	146	7%	Yes - territory boundaries
Spare-only overlap (no primary conflict)	1,418	70%	Yes - spare scheduling

Division-level overlap

Division	Active Segments	Multi-Grader %	Grading Visits	Context
Div 2	337	97%	100,147	Heaviest spare traffic
Div 7	367	95%	79,083	Heavy spare traffic
Div 5	286	69%	110,890	MG-048 territory + spares
Div 1	479	68%	99,236	MG-051 + spare coverage
Div 6	304	64%	152,960	Moderate
Div 3	386	60%	145,215	Moderate
Div 4	653	50%	172,385	Least overlap

What this means

The overlap problem is primarily a spare deployment coordination problem. The spares are doing useful work, but without coordinated scheduling they sometimes re-grade roads that the primary grader already completed.

4. Depot proximity and network coverage

Per-grader camp distance

Grader	Avg Start Distance	P90 Distance	Max Distance	Same Camp Return
MG-039 (Spare)	0.1 km	-	-	88%
MG-044 (Spare)	0.1 km	-	-	98%
MG-048 (Div 5)	0.1 km	-	-	99%
MG-050 (Div 4)	0.1 km	-	-	98%
MG-047 (Div 6)	0.5 km	-	-	100%
MG-046 (Div 3)	1.4 km	-	-	95%
MG-041 (Spare)	**3.2 km**	-	-	81%
MG-051 (Div 1)	**6.2 km**	-	-	96%

MG-051 is the clear depot outlier. That aligns with earlier GIS work showing Coaldale Shop sits well away from the demand centre of Division 1.

Network coverage

98.3% of gravel segments were visited - 2,500 of 2,542. The issue is not broad under-coverage. It is the fuel, time, and coordination cost of achieving that coverage.

Division	Segments	Visited	Coverage
Division 2	322	320	99.4%
Division 6	274	272	99.3%
Division 5	286	283	99.0%
Division 4	499	493	98.8%
Division 7	323	318	98.5%
Division 3	357	350	98.0%
Division 1	396	381	96.2%

5. Optimization opportunities ranked by impact

Tier 1: highest impact

Opportunity	Mechanism	Estimated Value	Timeline
Route optimization (PyVRP)	Optimized daily route sequencing within each division, minimizing deadhead travel and focusing on grading season.	$15,000-$35,000/yr fuel; $40,000-$90,000 fully loaded	Sprint 4 (June)
Spare deployment coordination	Assign spares to highest unmet demand and exclude already-graded segments from spare routes.	$5,000-$15,000/yr	Sprint 4
MG-048 investigation + resolution	Diagnose the 24% high-load anomaly. Savings could come from equipment fixes, route improvement, or both.	$5,000-$10,000/yr	Sprint 1-4

Tier 2: moderate impact

Opportunity	Mechanism	Estimated Value	Timeline
Division 1 depot staging	Use multi-day plans and overnight staging when MG-051 is working far west.	$3,000-$5,000/yr net	Sprint 4
MG-050/MG-048 swap	Place the most efficient grader on the heaviest division, subject to validating whether MG-048 load is machine-driven.	$3,500/yr	Pending investigation
Demand-driven scheduling	Replace fixed rotation with priority-based grading frequency.	Productivity gain	Sprint 4

Tier 3: compounding value

Opportunity	Mechanism	Estimated Value	Timeline
Weather-responsive scheduling	Grade after rain and defer during dry spells for better durability.	Longer road integrity	Phase 2-3
Grader Performance Score	Identify operators producing the most durable results and support peer learning.	Compounds over seasons	Sprint 5
Dust control coordination	Avoid grading dust-controlled roads after treatment and coordinate grading before calcium application.	$10,000-$25,000/yr avoided waste	Sprint 4
Treatment optimization	Allocate grading, dust control, and stabilization budgets segment by segment.	Capital allocation analysis	Phase 3

Combined 2026 projection

Lever	Conservative	Moderate	Optimistic
Route optimization (fuel)	$15,000	$25,000	$35,000
Spare coordination	$5,000	$10,000	$15,000
MG-048 resolution	$3,000	$5,000	$10,000
Depot staging (Div 1)	$2,000	$3,000	$5,000
Dust control coordination	$5,000	$10,000	$15,000
Visibility / behavior change	$3,000	$8,000	$15,000
2026 Total (fuel savings)	$33,000	$61,000	$95,000
Fully loaded (fuel + labor + depreciation)	$80,000	$150,000	$235,000

Planning takeaway

A moderate case points to about $61,000 in annual fuel savings and roughly $150,000 in fully loaded value when route optimization, spare coordination, mechanical follow-up, and visibility changes are combined.

6. Immediate recommendations

1. Share with Shawn Use Shawn's field perspective to validate spare deployment patterns and MG-048 operating conditions.	**2. Investigate MG-048** Ask Fleet to check hydraulic calibration, tire pressure, and blade setup against MG-050 as the comparison unit.
3. Approve dashboard deployment Publish the metrics on a weekly cadence so the fleet can track deadhead, overlap, and improvement against baseline.	**4. Approve the Route Planner pilot** Start with one willing division and measure route adherence, fuel use, and grading coverage after rollout.
5. Request the GIS priority layer The planner needs High/Medium/Low segment priority to generate demand-driven routes instead of fixed rotations.	**Bottom line** The fleet is covering the network. The opportunity is to cover it with less travel, less overlap, and better daily sequencing.

Final takeaway

The highest-value 2026 moves are route optimization, spare coordination, and solving the MG-048 anomaly. Everything else compounds on top of those three decisions.

All findings are derived from measured GPS breadcrumbs, verified fuel fill-up records, OEM telematics, and GIS road network data. No estimates or industry benchmarks - every number traces back to Lethbridge County data.